

Powered Lower-Limb Prostheses: Control, Estimation, Personalization, and Clinical Translation (IFAC WC 2026 — Open Invited Track Proposal)

Abstract

Powered lower-limb prostheses are transitioning from laboratory prototypes to clinically viable devices. This Open Invited Track (OIT) will convene the control, robotics, rehabilitation, and AI communities to accelerate robust **powered knee/ankle** solutions for transfemoral and transtibial amputees. We target advances in **model- and data-driven control, phase-variable and intent-aware estimation, user-specific personalization and co-adaptation, actuation and sensing, computational modeling and digital simulation for design and testing, and clinical translation** (benchmarks, datasets, trials, and outcomes). The track emphasizes cross-pollination among academia, clinics, and industry to help close the gap between state-of-the-art laboratory performance and reliable everyday use.

Keywords

Powered prosthesis; transfemoral; transtibial; powered knee–ankle; phase-variable control; variable impedance; intent recognition (EMG, inertial, vision, environment sensing); personalization & human-in-the-loop optimization; reinforcement learning; digital twins & simulation; safety & certification; open datasets; clinical trials; benchmarking.

IFAC Technical Committee for Review

Primary TC: 4.3 Robotics

Co-review (requested): 8.2 Biological and Medical Systems

Website

https://hurgroup.github.io/pages/ifac_wc2025_oit01

Detailed Description of the Topic

Motivation — Why this OIT now?

- **Clinical need & quality of life.** Individuals with transfemoral and transtibial amputations face elevated metabolic cost, fall risk, and limited mobility. Powered devices are programmable and can inject positive work, reduce compensations, and improve stability on stairs, ramps, and uneven terrain.
- **Convergence of technologies.** High torque–density actuators (quasi-direct drive, SEA/variable transmissions), compact batteries, high-rate sensing (IMUs, load cells, socket/interface sensing), and on-device compute enable real-time, adaptive control policies that are safer and more versatile.
- **Translation gap.** Despite impressive laboratory results, real-world robustness, personalization, and standardized evaluation remain open challenges. Adoption further requires **clinically explainable control**, repeatability, and validated outcomes.

What has been done

Representative advances include:

- **Phase-based locomotion control and autonomous task adaptation** for powered knee–ankle systems (e.g., speed/incline adaptation; swing–stance switching; stair ascent/descent).
- **Variable-impedance and hybrid control** that modulate joint impedance through the gait cycle; **model-based** and **data-driven** controllers that blend kinematic and impedance objectives.
- **Intent and environment awareness:** EMG-driven and pattern-recognition methods (residual musculature), gait-phase estimation from thigh/hip kinematics, and exteroceptive perception (vision/lidar) for upcoming terrain.
- **Clinical demonstrations and research platforms:** Utah Bionic Leg; Vanderbilt/UMich powered knee–ankle prototypes; commercial **Ottobock Empower** (powered ankle–foot) and **Össur Power Knee** (powered knee) that are catalyzing broader clinical evaluation.
- **Open-source and benchmarking efforts:** Emerging open designs, datasets, and evaluation protocols to facilitate reproducibility and side-by-side comparisons.

- **Modeling and simulation platforms:** Neuromusculoskeletal models, musculoskeletal simulations (e.g., OpenSim), and physics-based digital twins (e.g., MuJoCo, MyoSuite) are actively used for developing and testing the control strategies.

What's missing

- **Generalization:** Controllers that gracefully handle user- and task-variability across extended daily-living use (multi-speed, multi-incline, irregular terrain, perturbations).
- **Personalization at scale:** Fast, safe, and interpretable tuning that co-adapts device and user (onboard learning, preference-aware optimization), with **clinically meaningful parameters** and guardrails.
- **Shared autonomy & intent fusion:** Combining EMG, neuromechanics, inertial/force sensing, and environment perception with uncertainty-aware estimation and safety constraints.
- **Energy, weight, and comfort:** Co-design of actuators, transmissions, and sockets to reduce mass/power without sacrificing torque bandwidth or durability.
- **Robustness and Safety:** Robust control against internal and external faults and perturbations to ensure user safety.
- **Standardization:** Open benchmarks, common datasets, and multi-site clinical protocols that report comparable outcomes (metabolic cost, stability, mobility, long-term adherence).
- **Verification & certification:** Formal safety envelopes, runtime monitors, and risk management aligned with medical-device standards (e.g., IEC 60601-family) to accelerate clinical adoption.
- **Simulation-to-reality gap:** While simulation environments enable rapid prototyping and large-scale virtual testing, validated pipelines for transferring control strategies and personalization from simulation to hardware remain underdeveloped.

Call for Papers — Topics of interest

We welcome contributions (methods, systems, datasets, trials, negative results) including but not limited to:

1. **Control & Estimation**

- Autonomous, **phase-variable** control for knees/ankles; **variable impedance/hybrid** strategies; **MPC** and **safety filters** (e.g., CLF-CBF).
- **Intent recognition** from EMG/mechanical/vision; multimodal sensor fusion; adaptive state machines; disturbance observers for socket–ground interactions.
- **Learning in the loop:** Preference learning, policy adaptation, meta-/few-shot tuning, and robustness across subjects and tasks.

2. Hardware, Sensing & Co-Design

- High-torque, low-impedance actuation (quasi-direct drive, torque-sensitive transmissions, SEA/VSA); energy regeneration; battery and thermal management; co-design with ergonomic constraints (mass/inertia distribution, device packaging).
- Compact, reliable sensing in prosthetic limbs and sockets; haptics and sensory substitution for balance and foot placement; socket/interface ergonomics for stable sensing without discomfort.
- Ergonomic design and human–device interface: socket fit and comfort, load distribution, don/doff usability, skin integrity, and accommodation of anthropometric variability; methods to quantify and optimize ergonomics alongside control performance.

3. Human Factors & Clinical Translation

- **Personalization protocols** (clinically explainable), outcome metrics, and multi-center trials; safety certification pipelines; failures and field data.
- **Open datasets & benchmarks** (terrain-varying gait, sit–stand, stairs/ramps) and **prosthesis emulators** for rapid prototyping and comparative studies.

4. Applications & Case Studies

- Real-world deployment reports with longitudinal outcomes; comparative studies of powered vs. passive/microprocessor devices; rehabilitation integration.

5. Modeling & Simulation

- Neuromusculoskeletal modeling of amputee gait and prosthesis–user interaction.
- Digital twins and physics-based simulation for control design, tuning, and safety evaluation.
- Virtual prototyping of actuators, sockets, and user-specific adaptation schemes.

Relevance to IFAC

The track directly aligns with **TC 8.2** (biomedical systems, clinical studies, AI-in-healthcare) and **TC 4.3** (robotics: actuation, sensing, autonomy). It bridges modeling, identification, optimization, and control with translational evaluation and safety—core IFAC themes.

Organizers

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